Mark Scheme

PHYSICS · Paper 4 (A2)

A2 Mid-term Examination — 5 questions · 100 marks

Mark scheme — Question 1

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Question	Answer	Marks
1(a)	angle (subtended at centre of circle) when arc length = radius	**B1**
1(b)	$\omega = 2\pi / T$	**C1**
	$= 2\pi / (1.0 \times 60 \times 60)$ $= 1.7 \times 10^{-3}$ rad s^{-1}	**A1**
1(c)(i)	angle $= 1.7 \times 10^{-3} \times 1400$ $= 2.4$ rad	**A1**
1(c)(ii)	L = arc length / angle $= 0.44 / 2.4$ **or** $L = 0.44 \times (3600 / 1400) / 2\pi$	**C1**
	$L = 0.18$ m	**A1**
1(c)(iii)	$a = r\omega^2$	**C1**
	$= 0.18 \times (1.745 \times 10^{-3})^2$ $= 5.5 \times 10^{-7}$ m s^{-2}	**A1**
1(d)	centripetal acceleration is negligible compared with acceleration of free fall **or** numerical comparison establishing answer to **(c)(iii)** $\ll 9.81$	**B1**
	resultant force is negligible compared with weight (of modelling clay) (so variation is negligible) **or** force exerted by minute hand (approximately) equal (and opposite) to weight of modelling clay	**B1**

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Mark scheme — Question 2

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Question	Answer	Marks
1(a)	work done per unit mass	B1
	work (done on mass) moving mass from infinity (to the point)	B1
1(b)(i)	$-3.55 \times 10^7\ \text{J kg}^{-1}$	B1
1(b)(ii)	$\phi = -\frac{GM}{r}$ $M = -\frac{-3.55\times10^7\times4800000}{6.67\times10^{-11}}$ $= 2.55 \times 10^{24}\ \text{kg}$	B1
1(b)(iii)	$g = \frac{GM}{r^2}$ **or** $g = -\frac{\phi}{r}$	C1
	$= \frac{6.67\times10^{-11}\times2.55\times10^{24}}{4800000^2}$ **or** $= \frac{3.55\times10^7}{4800000}$ $= 7.4\ \text{N kg}^{-1}$	A1
1(b)(iv)	r in range 2.60×10^7 to 2.65×10^7 m	C1
	$\frac{mv^2}{r} = \frac{GMm}{r^2}$ **and** $v = \frac{2\pi r}{T}$ **or** $mr\omega^2 = \frac{GMm}{r^2}$ **and** $\omega = \frac{2\pi}{T}$	C1
	$T^2 = \frac{4\pi^2 r^3}{GM} = \frac{4\pi^2\times\left(2.65\times10^7\right)^3}{6.67\times10^{-11}\times2.55\times10^{24}} = 4.20 \times 10^9$	C1
	$T = 64\,800\ \text{s}$ $= 18\ \text{hours}$	A1

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Mark scheme — Question 2

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Question	Answer	Marks
1(c)	similarity – any one point from • inversely proportional to distance (from point) • points of equal potential lie on concentric spheres • zero at infinite distance	B1
	difference – any one point from • gravitational potential is (always) negative • electric potential can be positive or negative	B1

Question	Answer	Marks
2(a)	gas for which $pV \propto T$	M1
	where T is thermodynamic temperature	A1
2(b)(i)	evidence of two temperature conversions between °C and K	B1
	two calculations shown, one for each state e.g. $\frac{1.10\times10^{5}\times540\times10^{-6}}{(273+27)} = 0.198$ **and** $\frac{6.70\times10^{6}\times30\times10^{-6}}{(273+742)} = 0.198$	A1
2(b)(ii)	work is done on the gas	M1
	internal energy increases (so temperature increases)	A1

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Mark scheme — Question 3

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Question	Answer	Marks
2(a)(i)	work done per unit mass	B1
	work (done) moving mass from infinity (to the point)	B1
2(a)(ii)	$\phi = -GM/r$ $= -(6.67 \times 10^{-11} \times 7.3 \times 10^{22}) / (1.7 \times 10^{6})$	C1
	$= -2.9 \times 10^{6}\ \text{J kg}^{-1}$	A1
2(b)(i)	$E_P = m\phi$	B1
2(b)(ii)	$\frac{1}{2}mv^2 + m\phi = 0$	M1
	correct algebra leading to $v = \sqrt{(-2\phi)}$	A1
2(c)	speed $= \sqrt{(2 \times 2.9 \times 10^{6})}$ $= 2400\ \text{m s}^{-1}$	A1
2(d)	$\frac{1}{2}m\langle c^2\rangle = (3/2)kT$	C1
	$3.34 \times 10^{-27} \times \langle c^2\rangle = 3 \times 1.38 \times 10^{-23} \times 400$	C1
	$c_{\text{r.m.s.}} = 2200\ \text{m s}^{-1}$	A1
2(e)	r.m.s. speed is an average so many molecules have speeds greater than the escape speed **or** there is a distribution of molecular speeds (around the r.m.s. value) so many molecules have speeds greater than the escape speed	B1

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Mark scheme — Question 4

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Question	Answer	Marks
3(a)	P: total energy Q: potential energy R: kinetic energy	B2
3(b)	$E = \frac{1}{2}m\omega^2 x_0^2$ **or** $E = \frac{1}{2}mv_0^2$ **and** $v_0 = \omega x_0$	C1
	$6.4\times10^{-3} = \frac{1}{2}\times0.130\times\omega^2\times0.015^2$ $(\omega^2 = 438)$ $(\omega = 20.9)$	C1
	$T = 2\pi / \omega$	C1
	$= 2\pi / 20.9$ $= 0.30$ s	A1
3(c)(i)	resistive forces	B1
3(c)(ii)	0.92^6	C1
	decrease in energy $= 6.4 - (6.4 \times 0.92^6)$ $= 2.5$ mJ	A1
3(c)(iii)	light damping because the amplitude of oscillations gradually reduces **or** light damping because the system still oscillates	B1

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Mark scheme — Question 5

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Question	Answer	Marks
3(a)	$a = -\omega^2 x$	M1
	a = acceleration, x = displacement from equilibrium position and ω = angular frequency	A1
3(b)(i)	$x_0 = 0.12$ m	A1
3(b)(ii)	$v = \omega\sqrt{(x_0^2 - x^2)}$ two (x, v) pairs correctly read from Fig. 3.2 (one may be (x_0, 0) or value of x_0 from **(i)**)	C1
	e.g. $0.20 = \omega\sqrt{(0.12^2 - 0)}$ leading to $\omega = 1.7$ rad s^{-1}	A1
3(b)(iii)	$E = \frac{1}{2}M\omega^2 x_0^2$	C1
	$0.050 = \frac{1}{2} \times M \times 1.67^2 \times 0.12^2$ $M = 2.5$ kg	A1
	or	
	$(E_K)_{max} = \frac{1}{2}Mv_0^2$	(C1)
	$0.050 = \frac{1}{2} M \times 0.20^2$ $M = 2.5$ kg	(A1)
3(c)(i)	loss of (total) energy (of system)	B1
	due to resistive forces	B1
3(c)(ii)	closed loop surrounding the origin with maximum x at ± 0.060 m passing through $v = 0$	B1
	maximum velocity shown as ± 0.10 m s^{-1} passing through $x = 0$	B1

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